Predictive Pricing Model for Real Estate Packs

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1) Context & Use Case

Leboncoin sells advertising packs (10 to 200 ads/month) to professional real estate agencies.

Agencies can either buy packs (monthly allowance of ads), use FIM/manual uploads (pay-per-ad), and buy premium options (boost, top results, phı

The goal is to set prices that grow revenue while keeping churn and downsell under control and encouraging migration from manual to packs.

2) Business Goals

- Revenue growth: +5% portfolio uplift (≤ +18% per product)
- Churn guardrail: ≤5% (natural baseline ≈5%)
- Pack mix: reduce downsell, enable upsell if revenue-positive
- Migration: move agencies from FIM/manual into packs
- Volume: maintain or grow total ads published

3) Data & Features

We used four years of renewal-level data (~10k agencies). Representative features include:

- Prices: pack prices, FIM price, premium price (net of discount)
- Current state: {Manual, Pack10, Pack20, Pack50, Pack100, Pack200}
- Usage: ads published, pack fill rate, number of mandates
- Spend: premium & boutique spend, total spend history
- Profile: tenure, discount history, agency size
- Market: competitor presence (Seloger, Bien'ici, MeilleursAgents), region
- Budget: affordability = log1p(total_spend); budget_pressure = total_spend / peak_spend_12m

Modeling format: long (agency × renewal × alternative). One row per candidate option (Manual, each Pack size, Churn) per renewal event.

Data Snapshot (Clean two-row records per agency)

agency	period	current_state	choice	price_fim	price_10	price_20	pri
A	t1	Pack20	Pack50	7.0	50	100	200

ExpAds	8
Premium Spend	20
Region	North
Derived	short = max(0, 8-10) = 0; slack = max(0, 10-8) = 2

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Reshaped (Long Format — alternatives per renewal)

agency	period	alt	chosen	price_alt	slots_alt
A	t1	Pack20	0	100	20

Inertia	1 (current = Pack20)
Short	8
Slack	0
FIM Cost Proxy	–
ExpAds	28
Region	South

agency	period	alt	chosen	price_alt	slots_alt
A	t1	Pack50	1	200	50

Inertia	0
Short	0
Slack	22
FIM Cost Proxy	–
ExpAds	28
Region	South

agency	period	alt	chosen	price_alt	slots_alt
A	t1	Manual	0	7.0	–

Inertia	0
FIM Cost Proxy	7.0 × 28 = 196
ExpAds	28
Region	South

agency	period	alt	chosen	price_alt	slots_alt
B	t1	Manual	0	7.0	–

agency	period	alt	chosen	price_alt	slots_alt

Inertia	1 (current = Manual)
FIM Cost Proxy	7.0 × 8 = 56
ExpAds	8
Region	North

agency	period	alt	chosen	price_alt	slots_alt
B	t1	Pack10	1	50	10

Inertia	0
Short	0
Slack	2
FIM Cost Proxy	–
ExpAds	8
Region	North

3. Modeling Approach

We use a multinomial logit (MNL) to predict renewal choices across alternatives: {Manual, Pack10, Pack20, Pack50, Pack100, Pack200, Churn}.

Utility specification

Manual:

$$U_{\text{manual},i} = \alpha_{\text{manual}} - \beta_{\text{fim}} \left(\text{price}_{\text{fim}} \cdot \text{ExpAds}_i\right) + \beta_{\text{inertia}} \cdot \mathbf{1}\{\text{current} = \text{Manual}\} + \Gamma^\top X_i + \text{Region}_i.$$

Pack j:

$$U_{j,i} = \alpha_j - \beta_{\text{price}} \cdot \text{price}_j - \beta_{\text{short}} \cdot \max(0, \text{ExpAds}_i - \text{Slots}_j) - \beta_{\text{slack}} \cdot \max(0, \text{Slots}_j - \text{ExpAds}_i) + \beta_{\text{inertia}} \cdot \mathbf{1}\{\text{current} = j\} + \Gamma^\top$$

Churn:

$$U_{\text{churn},i} = \alpha_{\text{churn}} + \beta_{\text{stress}} \cdot \mathbf{1}\{\%\Delta\text{price}_{\text{current}}\ \text{high}\} + \tilde{\Gamma}^\top \tilde{X}_i + \text{Region}_i.$$

From utilities to probabilities

$$\hat{p}_{i,k} = \frac{\exp(U_{i,k})}{\sum_m \exp(U_{i,m})}.$$

4. Training

- Unit: agency × renewal
- Target: observed choice
- Format: long (1 row per alternative)
- Estimation: maximum likelihood (log-likelihood)
- Regularization: sign constraints for interpretability (β_price > 0, β_inertia > 0)

5. Parameter Interpretations

- α_j: baseline attractiveness of each alternative
- β_price: sensitivity to pack price
- β_fim: penalty from manual FIM cost
- β_short: penalty if expected ads exceed slots (shortage → upsell pressure)
- β_slack: penalty if slots exceed expected ads (slack → downsell pressure)
- β_inertia: status-quo bias
- β_stress: churn trigger when price hike is high

- Γ: customer profile effects (usage, spend, tenure, competitor presence, affordability)

6. Results (Illustrative)

We simulate price changes (≤ +18%) and recompute utilities → choice probabilities → expected revenues.

New Prices Example

- Packs +12% → P10=€22.40, P20=€39.20, P50=€89.60
- FIM +18% → €6 → €7.08
- Premium +8% → €20 → €21.60

Agency A1 — Capacity-Constrained

Current	Pack20, ExpAds=28, fill=92%, premium high, competitor=1
Probabilities	Pack50=0.55, Pack20=0.35, Pack10=0.03, Manual=0.02, Churn=0.05
Revenues (€/mo)	Pack20≈268.6, Pack50≈265.1
Expected Revenue	≈ €266–267
Conclusion	Upgrade to Pack50 likely (shortage penalty).

Agency A2 — Manual Mid-Volume

Current	Manual, ExpAds=12, low premium, no competitor
Probabilities	Pack10=0.60, Manual=0.25, Pack20=0.10, Churn=0.05
Revenues (€/mo)	Manual≈106.6, Pack10≈58.2, Pack20≈60.8
Expected Revenue	≈ €70–75
Conclusion	Migration to Pack10 likely (stability, retention).

Agency A3 — Under-Utilized, Price-Sensitive

Current	Pack50, ExpAds=18, discount history=1, competitor=1
Probabilities	Pack20=0.40, Pack50=0.30, Pack10=0.15, Manual=0.10, Churn=0.05
Revenues (€/mo)	Pack50≈130–135, Pack20≈80–85, Manual≈155–160
Expected Revenue	≈ €110–120
Conclusion	Likely downgrade to Pack20; churn risk present.

> Aggregate (toy; 3 agencies): Mix shifts A1→Pack50, A2→Pack10, A3→Pack20; churn ≈ 5%; ads published net positive overall.

6b) Model Evaluation & Backtesting (Cost-Sensitive)

Why not plain accuracy?

Predicting *upsell* when reality is *downsell* is far costlier than the reverse. We evaluate with asymmetric business costs.

Movement labels

We map choices to movements relative to current state: UP, SAME, DOWN, MANUAL, CHURN. For agency *i*, the observed movement is y_i and th probabilities $\hat{\mathbf{p}}_i$.

Asymmetric cost matrix

Rows = truth, columns = decision:

Truth \ Pred	UP	SAME	DOWN	MANUAL	CHURN
UP	0	3	4	5	6
SAME	6	0	3	4	5
DOWN	**10**	4	0	5	6

Truth \ Pred	UP	SAME	DOWN	MANUAL	CHURN
MANUAL	7	4	5	0	5
CHURN	8	5	6	5	0

Risk-minimizing decision rule

Pick the decision k that minimizes expected cost under $\hat{\mathbf{p}}_i$:

$$\hat{y}_i = \arg\min_k \sum_j \hat{p}_{i,j}\, C_{j,k}.$$

Primary metric: CostScore

Average expected cost for the model vs. a naive baseline (always predict SAME):

$$\overline{C}_{\text{model}} = \frac{1}{N}\sum_{i=1}^{N}\sum_j \hat{p}_{i,j}\, C_{j,\hat{y}_i}, \qquad \text{CostScore} = 1 - \frac{\overline{C}_{\text{model}}}{\overline{C}_{\text{base}}} \in [0,1].$$

Secondary diagnostics

- **Protective recall** on {DOWN, MANUAL, CHURN}
- **Upside precision** on UP
- **Top-1 accuracy** (plain, for interpretability)
- **Log-loss** (probability calibration): $-\frac{1}{N}\sum_i \log \hat{p}_{i,y_i}$

Tiny numerical example

- **Setup:** Single Pack20 agency, ExpAds = 28; prices: Pack10 = 50, Pack20 = 100, Pack50 = 200; FIM = 7/ad.
- **Parameters (toy):** β_price = 0.02, β_short = 0.10, β_slack = 0.03, β_fim = 0.02, β_inertia = 0.8; intercepts: α_man = –0.5, α_churn = –3, α
- **Utilities (rounded):** U(UP) ≈ –4.66; U(SAME) ≈ –2.00; U(DOWN) ≈ –2.80; U(MANUAL) ≈ –4.42; U(CHURN) = –3.00.
- **Probabilities (softmax):** {UP: 0.035, SAME: 0.506, DOWN: 0.227, MANUAL: 0.045, CHURN: 0.186}.
- **Expected cost by decision:**
  - E[C|UP] ≈ 7.11
  - E[C|SAME] ≈ **2.13** ← lowest
  - E[C|DOWN] ≈ 3.00
  - E[C|MANUAL] ≈ 4.27
  - E[C|CHURN] ≈ 4.33

  Decision: **SAME** (risk-minimizing).

Offline backtest (out-of-sample)

- Population: ~17k customers (≈7k Manual, ≈10k Packs)
- Baseline (always SAME): CostScore = 0.00
- Model (MNL + risk-min decision): CostScore ≈ **0.42**; Protective Recall ≈ **0.74**; Upside Precision ≈ **0.68**; Top-1 Accuracy ≈ **0.62**; Log-Loss ≈ 16
  baseline

> **Interpretation.** The risk-aware decision rule prevents costly UP↔DOWN mistakes by acting conservatively when probabilities are diffuse; residu
> occur across adjacent pack sizes (lower business cost).

7. Deployment Plan

- Simulation engine: recompute utilities/probabilities under new prices
- Scenario testing: revenue, churn, upsell/downsells, ad volume
- Guardrails: +18% price cap, churn ≤5%
- Regional calibration: local sensitivities
- Monitoring: compare predicted vs. actual renewals

8. Business Impact

- Projected +5–7% revenue uplift within constraints
- Churn maintained at ≈5%

- Increased migration from manual → packs (more inventory)
- Stakeholder confidence in pricing rollout

9. Key Learnings

- Renewal behavior explained by inertia, fit (short/slack), and price sensitivity
- Manual→Pack migration driven by FIM vs Pack cost trade-off
- Non-adjacent jumps triggered by shortage penalties
- Churn risk concentrated in discount-history + competitor-present segments

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